Potential-Guided Planning in Magic Tower

A Deterministic Long-Horizon Resource-Allocation Benchmark

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Problem: deterministic but hard

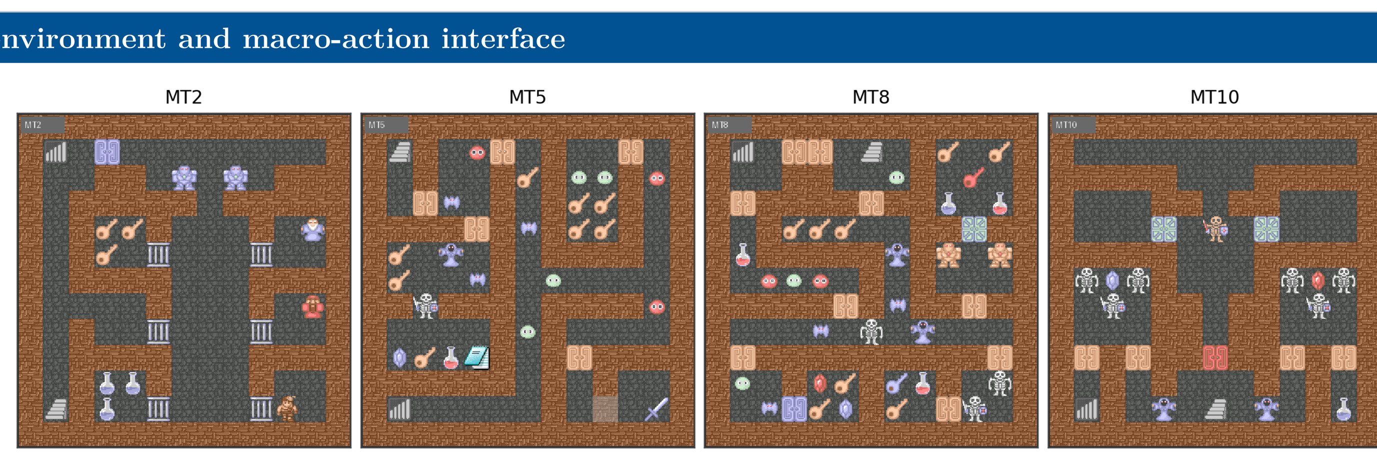
Magic Tower is a fully observable single-agent RPG puzzle. Its difficulty is not stochasticity, but irreversible resource scheduling over HP, ATK, DEF, keys, money, doors, items, events, and enemies.

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}_{\text{valid}}, f, r_{\text{task}}, \gamma, \rho_0, \mathcal{G}, \mathcal{F}, H).$$

$$V^*(s) = \max_{a \in \mathcal{A}_{\text{valid}}(s)} \left[r(s,a) + \gamma V^*(f(s,a)) \right].$$

- Local feasibility is misleading: a legal action can destroy a future route.
- Combat is threshold-sensitive; one ATK/DEF point can change many future losses.
- Terminal reward is sparse; random exploration rarely observes success.

Environment and macro-action interface



We use a deterministic simulator and macro-actions over interaction nodes. Ordinary movement is hidden inside the transition function, while resource-changing decisions remain explicit.

$$\mathcal{A}_{\text{macro}}(s) = \{a_i : i \in I_f, i \in R_f(s)\} \cup \{a_j : j \in \partial R_f(s), L(s,j) = 1\}.$$

Reachability is accelerated using static floor topology, interaction-node tables, blocker bitsets, cached BFS, and lazy path reconstruction.

Potential-based reward shaping

Reference routes estimate a stage-conditioned potential, not a replacement for terminal evaluation:

$$r_{\text{search}}(s,g,a,s',g') = r_{\text{task}}(s,a,s') + \lambda \big[\gamma \Phi_\phi(s',g') - \Phi_\phi(s,g) \big].$$

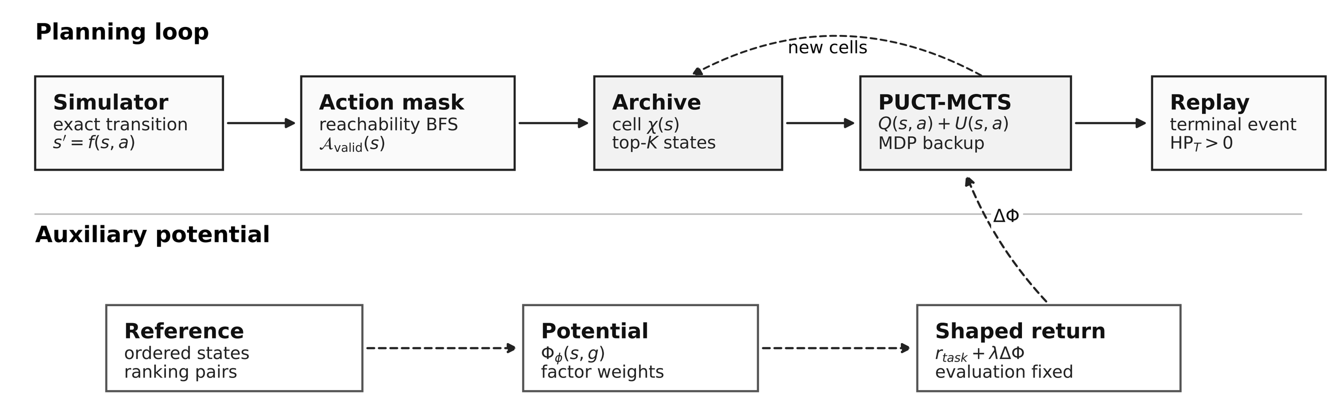
$$\Phi_\phi(s,g) = \sum_k w_{g,k} c_k(s;g) + \sum_k w_k^{\text{global}} c_k(s;g).$$

Factors include hero resources, key pressure, reachable and blocked resources, enemy damage, ATK/DEF threshold drops, stage progress, and terminal HP margin.

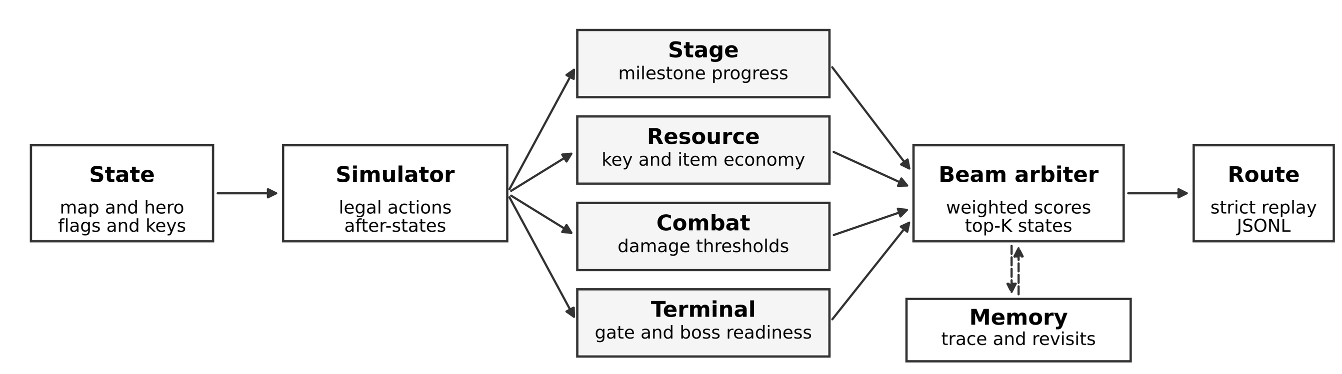
Three complementary approaches

1. **Reference-route construction**: audited trajectories validate simulator semantics and expose feasible resource schedules.
2. **AlphaGo Zero-style planning**: graph policy/value learning, single-agent PUCT-MCTS, and potential-shaped returns guide legal action search.
3. **Multi-agent reference planner**: specialized modules score the same legal candidates; a beam arbiter combines stage, resource, combat, and terminal-objective reasoning.

Planning system



Solid arrows denote simulator/search data flow. Dashed arrows denote auxiliary shaping; strict replay remains the success criterion.



All decisions are made over simulator-verified legal candidates; reference alignment is used only in this planner.

Policy/value model and PUCT

The graph policy/value model encodes interaction nodes and outputs a masked action prior and scalar value:

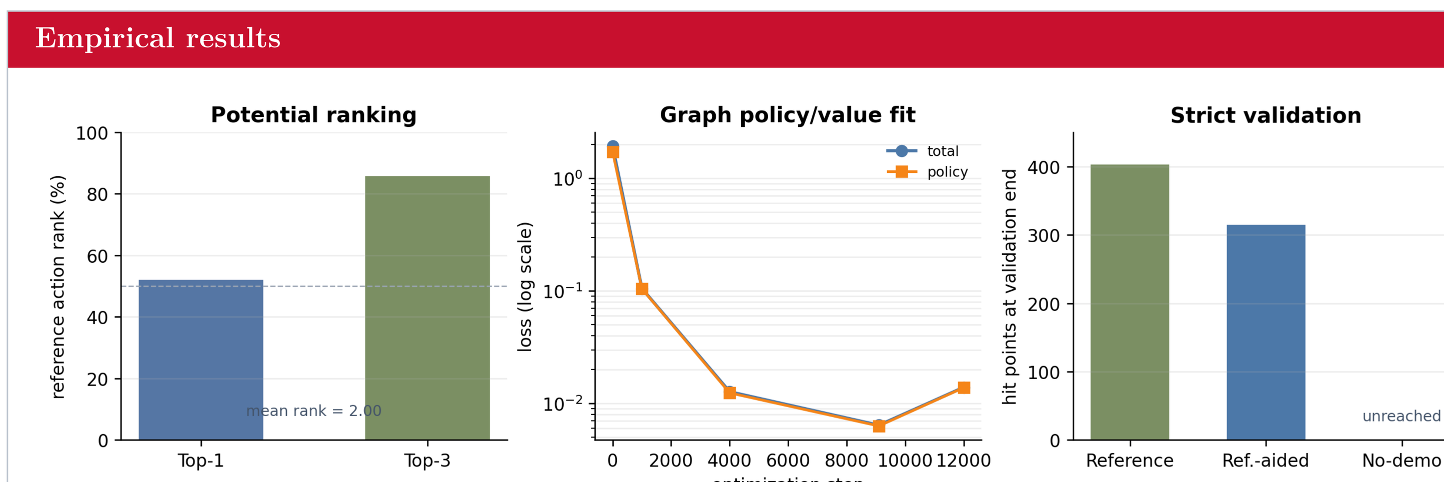
$$f_\theta(G_s, g) = \left(p_\theta(\cdot \mid s, g), v_\theta(s,g) \right).$$

PUCT selection in the single-agent setting uses

$$a = \arg \max_{a \in \mathcal{A}_{\text{valid}}(s)} \left[Q(s,a) + c_{\text{puct}} P(s,a) \frac{\sqrt{\sum_b N(s,b)}}{1 + N(s,a)} \right].$$

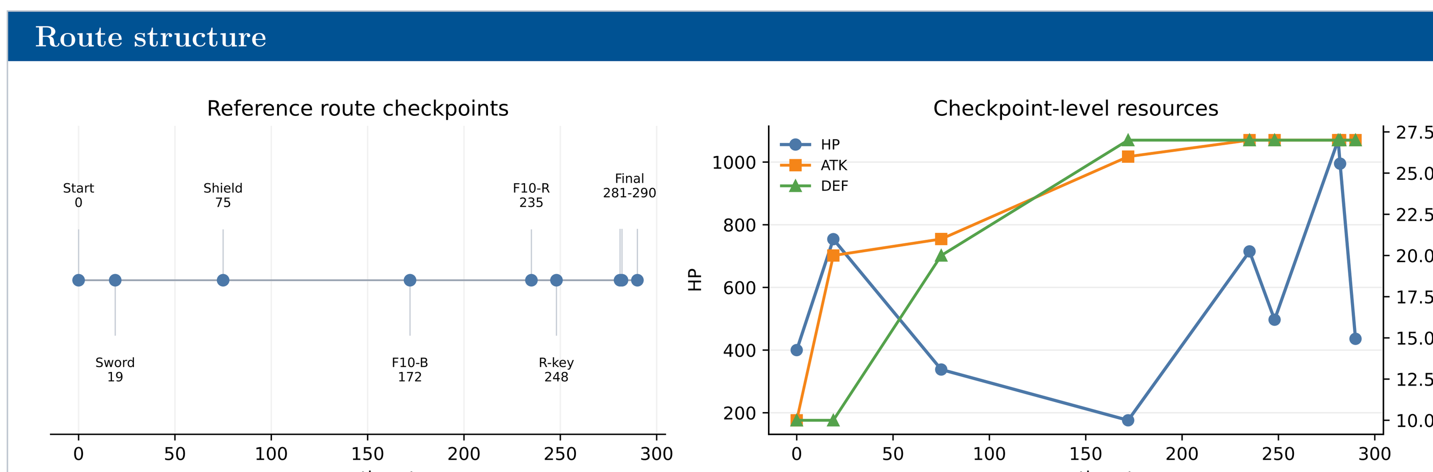
Backups accumulate deterministic task return over a max tree.

Empirical results



Condition	Signal	Outcome
Reference validation	strict replay	terminal success, $\mathrm{HP}_T = 403$
Potential shaping	ranking	Rank@1 52.1%, Rank@3 85.8%
Policy/value search	PUCT	terminal success, $\mathrm{HP}_T = 315$
Multi-agent planner	beam search	291 macro-actions, $\mathrm{HP}_T = 436$
No-demonstration archive	exploration	milestones reached, terminal incomplete

Route structure



The multi-agent route reaches the sword at step 19, shield at step 75, Floor-10 resource milestones before the terminal stage, red key at step 248, and terminal success at step 290. It deliberately trades HP for ATK/DEF thresholds, then rebuilds the terminal HP margin.

Benchmark outlook

Magic Tower can become a benchmark for long-horizon LLM and multimodal agents:

- synchronized symbolic JSON, natural-language summaries, and video frames;
- simulator-grounded actions with strict deterministic replay;
- thousands of community tower instances and many walkthrough episodes;
- tracks for trajectory modeling, offline RL, reward learning, and held-out map generalization.

The multi-agent planner gives a minimal protocol for coordinated agent reasoning over a shared candidate-action table.